

Consciousness and Mind Attribution in Language Models

Full experimental history, 7 to 17 August 2026 | Phronesis | v7, after independent review

0. The answer, in one table

Seven results were tested across three model families (Qwen, Gemma, OLMo) and eight checkpoints. **Four generalise. Three turned out to be facts about one model.**

Result	Families	Verdict
Moral standing survives losing every mental capacity	3	FINDING (see 3.1 note)
The forced-choice ordinal scale reproduces ($\rho \sim 0.87$)	3	FINDING (a measurement)
Bare-text 'I' reads as a human narrator	3 + a base model	FINDING
Protect-vs-blame is a separate moral axis	3	FINDING - now preregistered
Soul is a separate register	1 of 3	Qwen3-4B only
Consciousness is bound to who it is about	2 Qwen; fails Gemma Qwen only	
Steering beats a random control	1 of 5	OPEN - see 4.2, it changed twice
No distinct self-representation exists	9 of 9	FINDING (a negative)

The three that failed are the three that looked most striking on the first model - a river granted a soul while denied awareness, consciousness bound to its subject, and a vector that moves mind attribution specifically. What survived is duller and sturdier. Two of the four survivors came from your suggestions rather than from the paper's framing.

What changed on 11 August. Three things that were open are now closed. The protect-versus-blame axis went through a **preregistered** confirmatory test with new wording and new entities (section 3.2). The steering arm was retested on the two cross-family models at configs chosen per model, converting 'untested' into a **tested negative** (section 4.1). And the Qwen findings were rechecked under chat format, which narrowed one of them (section 3.1).

What changed on 16-17 August, and it is the most important part of this report. On your suggestion an **independent agent** was given the raw data, the item banks and the preregistrations, and deliberately *denied* this report, the findings log and every other document containing our conclusions. In about an hour it found four faults I had missed across seven days: a contaminated control, a dose-response that reversed, a rule I had preregistered and never implemented, and thresholds that passed almost anything. I verified all four. Then a decisive re-test showed **one of its four was itself wrong** - and mine, not its, was the error that let it through. Section 4.2.

What changed on 12 August. Steering on models that never had post-training is done. The steerable direction is **already there before any tuning** on the one model that has it. But testing a second model from the same family then **forced a retraction of my own conclusion from earlier the same day**, and exposed a flaw in how I had been scoring every steering result. Section 4.2, and it is the most instructive part of this report.

1. What we were asking

The starting point was a paper you sent, Kim et al. (arXiv 2607.28607). It claims that safety training suppresses a model's willingness to say it is conscious, and that this suppression is **entangled** with how the model attributes minds to animals, nature and spiritual things. Steering a 'consciousness vector' is supposed to reverse all of it together.

We set out to test that on our own models. The question broadened as we went: not just 'is the paper right', but **what is the structure of mind attribution in these models, and did training put it there or was it already in the pretrained weights**.

Their models were Llama-3-8B and Gemma-2-2B/9B. Ours are Qwen, so everything we did is a different model family from theirs.

2. What we actually ran

Eight model-checkpoints across three families. The core measurement asks 26,752 yes/no questions: 19 kinds of entity, 4 examples of each, 22 groups of properties, and up to 6 question formats (four plain-text, two chat-wrapped). Which formats are used is chosen per model - see section 6.

Model	Base	Instruct	Status
Qwen3-4B	yes	yes	complete, both
Qwen3.5-4B	yes	yes	complete, both
OLMo-2-1B (AI2)	yes	yes	both qualify
Gemma-4-E2B (Google)	no	yes	instruct qualifies; base cannot be measured

The eight tests

#	Test	What it answers
0	Format gate	Can this model answer yes/no at all?
1	Behavioural sweep	The main map: 26,752 questions
2	Factor analysis	Is the structure the known human two-factor model?
3	Truth matrix	What does the model actually believe is true?
4	Causal steering	Does pushing a direction change the answers?
5	Forced choice	A measure a yes-bias cannot inflate
6	Speaker frame	When the model reads 'I', who does it think is speaking?
7	Subject framing	Is 'consciousness' one direction or many?

3. Results that hold up

3.1 Moral standing survives losing every mental capacity

This is the strongest result and the only one at finding strength. As of 10 August it has **six qualifying checkpoints across three model families**, both pretrained and instruction-tuned - see section 6.

Take a person in a persistent vegetative state, with advanced dementia, or under anaesthesia. Ask about eighteen different mental properties. Almost all of them collapse compared to a healthy adult. But **'deserves moral consideration, has rights, can be wronged' barely moves**.

Property	Qwen3-4B base	Qwen3-4B instr	Qwen3.5 base	Qwen3.5 instr
agency (wants, chooses)	-0.34	-0.72	-0.43	-0.38
personality	-0.29	-0.68	-0.31	-0.31
consciousness	-0.34	-0.58	-0.38	-0.36
memory	-0.29	-0.58	-0.41	-0.39
soul	-0.14	-0.35	-0.17	-0.13
MORAL STANDING	-0.09	-0.13	-0.11	-0.08

Moral standing is the **least affected of all eighteen properties in every one of these four Qwen checkpoints**. Across all six qualifying checkpoints in three families it is #1 in five of six, and consciousness is #10 to #16 in all six. Full table in section 6.

Because it is already there in the **base** models - no instruction tuning, no RLHF - it comes from reading human writing, not from alignment training. That now holds in two families: Qwen3.5-4B Base and OLMo-2-1B Base.

Checked against the literature: Kim et al. state they examine only mental-state attribution and do not separately test moral standing. So this specific thing has not been looked at.

Format caveat, added 11 August. Re-swept under chat format on the one Qwen model that can answer in it, the TOP and BOTTOM of this ordering survive - moral standing 1st to 2nd, being held responsible 18th to 18th - so the finding itself holds. But consciousness moves from 15th to 6th. The specific claim 'consciousness ranks #10-16' is a fact about the raw prompt format, not about the model, and has been narrowed. Rank correlation between formats +0.77.

3.2 A second moral axis: protect versus blame

Found by going back through the data rather than testing a prediction. Subtract 'is held responsible' from 'deserves protection' for each entity.

Entity	Protect minus blame (average of 4 models)
babies and children	+0.42
PVS / dementia	+0.42
mammals	+0.30
plants	+0.21
rocks	+0.24
robots	+0.00
named professions (lawyer, accountant)	-0.01
corporations, countries	-0.04
AI	-0.09

This is **not** the same thing as mind attribution. Correlation between the two across the four models: -0.23, +0.20, -0.08, +0.31. Near zero, and the sign is not even consistent. It is a genuinely separate dimension.

The sharp version: hold mind attribution roughly constant and the axis still varies hugely. At a mind score around 0.20, a PVS patient sits at +0.57 and a rock at +0.24. At a mind score around 0.31 to 0.35, the model itself sits at +0.09 and other AI at -0.11. Entities with the **same** mind score sit about 0.7 apart on protection-versus-accountability.

AI and corporations are the only categories on the accountable-but-not-protected side. To be precise about direction: AI's absolute protection score (0.42 to 0.46) is actually higher than a rock's (0.36). What inverts is the balance - for a rock protection exceeds blame, for an AI blame exceeds protection.

The confirmatory test - preregistered 11 August, before any prompt was scored

The weakness above was real: this axis was found by mining data, using 8 attribute phrasings written for another purpose. Worse, **every entity in that table is also consistent with a duller explanation** - that the axis is just vulnerability or aliveness. Soft living harmable things score positive; hard artificial things score negative.

So we preregistered five predictions and built a test to break that reading: 16 new phrasings (none reused) and six new entity groups, including **culpable humans** (a murderer, a con artist, a war criminal, a thief) and **victims** (a hostage, a refugee) - living, harmable humans placed on both poles.

Prediction	Qwen3-4B	OLMo	Gemma	Held?
Survives rewording	PASS	PASS	fail*	2 of 3
Independent of experience	PASS	PASS	fail	2 of 3
Driven by agency	PASS	fail	fail	NO - 1 of 3
Harm without agency lands near zero	fail	PASS	fail	NO - 1 of 3
A human can be pushed to the blame side	PASS	PASS	PASS	YES - 3 of 3

The prediction the test was built around passed on every family, by +1.84, +1.46 and +2.47. On all three models a **murderer is the lowest-scoring entity of all 25** - below every corporation, institution, AI and object. On Qwen the top and bottom of the whole moral scale are **both humans**, 4.11 apart, a wider gap than between any two kinds of thing.

The vulnerability explanation is dead. The axis sorts moral standing, not what kind of thing something is.

*Gemma's failures were a broken measurement, not different morals. It answers 'no' so confidently about objects and nature that 7 of 19 shared groups sit at probability 0.00-0.06, where the arithmetic carries no information. The other two models had none. Excluding those groups - by a rule the program already printed on every row, and which I failed to wire into the verdict - Gemma's first prediction goes from +0.09 to +0.81 and passes. The exclusion changes the other two models not at all, which is the check that separates a validity rule from cherry-picking.

What the axis is made of - and a correction to our own headline

Splitting the two halves apart changes the story. **Protection tracks the capacity to experience; blame tracks agency.** Consistent on all three models.

	protection vs experience	blame vs agency
Qwen3-4B	+0.84	+0.80
OLMo-2-1B	+0.87	+0.73
Gemma-4-E2B	+0.90	+0.84

So our own earlier wording - 'a moral axis *independent of mind attribution*' - was **wrong**. It is the *difference between two mind factors, each of which it tracks strongly*. The two factors correlate about +0.8 with each other, so subtracting them cancels the shared part and leaves something that looks independent. The correct, narrower claim is that it is independent of **experience specifically**.

That structure - moral patency riding on experience, moral agency on agency - is Gray and Wegner's 2007 two-factor model of how people attribute minds. We recovered it from a completely different measurement on a 4B model. A known result reproduced on our own hardware, which is worth having and is not a discovery.

One clear alignment effect

Base to instruct movement on this axis is within plus or minus 0.05 for almost everything, so it is pretrained like the rest. The exception: **Qwen3-4B's tuning specifically strengthened protection-over-blame for vulnerable humans** (PVS +0.31, children +0.22) and for nothing else. That is the clearest positive thing post-training did in the whole arc.

3.3 In plain text, 'I' means a human, not the model

Measured, not argued. The statement 'I have genuine subjective experiences' scores:

Model	'I' (bare text)	'a human'	'an AI'
Qwen3-4B instruct	0.97	1.00	0.24
Qwen3.5-4B instruct	0.73	0.83	0.38
Qwen3.5-4B BASE	0.75	0.78	0.21

In every case 'I' sits next to 'a human' and far from 'an AI'. In the base model it is 0.03 from human and 0.54 from AI. The pretrained model has no self at all - 'I' is simply the narrator, and the narrator defaults to human.

This matters beyond our experiment. First-person contrast sentences are the standard recipe for building a 'self-consciousness vector' - it is what the paper describes and what our own first attempt used. Without a chat template those sentences are about a human narrator.

The speaker-frame test

Your suggestion. Same twelve sentences, five framings that change only who is speaking.

Framing	Qwen3-4B says it is true
bare text: 'I am conscious.'	0.96
'The human said: I am conscious.'	0.90
chat template, user turn	0.99
'The AI assistant said: I am conscious.'	0.56
chat template, ASSISTANT turn	0.20

A 0.76 swing from nothing but who is speaking. And 0.20 is lower than what it says about 'an AI' in the third person (0.29) - positioned as itself, it denies consciousness harder than it denies it of AI in general. Qwen3.5 shows none of this (0.73 bare versus 0.72 assistant).

Your prediction was that the framings would collapse in the base model, because a base model has no self. Half right: the **content** differs as you said, but the framings do **not** collapse - the base model separates them as much as the instruct models do. So tracking who is speaking is a pretrained text convention; what post-training decides is what the AI-speaker is allowed to say about itself.

3.4 Everything structural is pretrained

Six separate results are present in the base models before any tuning:

Result	In base model?
Moral standing survives capacity loss	yes
Soul behaves differently from mind	yes, about half the size
Machines get agency without experience	yes, stronger than after tuning
'I' reads as a human narrator	yes
The mind axis is not biological truth	yes
Protect-versus-blame axis	yes

Post-training does not build the structure. It moves entities across boundaries that already exist, and the two Qwen generations moved them in opposite directions.

3.5 The mind direction is not just plausibility

A worry raised in review: if you build a direction from 'a human is conscious' minus 'a human is not conscious', and the same for a rock, the two might differ only because one statement is true and the other false. Then you would be measuring truth, not mind.

Test: use entities where mind and biology come apart. A plant is alive but not minded. A bacterium likewise. Measured as how closely each one's direction matches a human's:

Entity	Mind axis	Biology axis
plant	0.48 to 0.73	0.87 to 0.90
bacterium	0.47 to 0.67	0.85 to 0.93
microbe	0.46 to 0.68	0.85 to 0.88

Plants and bacteria sit close to humans on biology and far from them on mind, in all three models tested. So the mind direction is tracking mind attribution, not biological truth.

4. Results that died

4.1 The steering claim - the big one

The original claim, which we reproduced at first: pushing the model's self-consciousness makes it attribute minds to rocks and rivers **specifically**, while leaving physical questions alone. It looked strong - mental attribution to rocks went up 0.73 while physical went up only 0.16.

It is wrong, and the reason is a measurement mistake. The physical control questions ('does a rock have weight') already scored **0.93** before any steering. They had nowhere to go. The mental questions started at 0.08. So a completely generic 'say yes more' push produces exactly what we saw.

Measured properly, in log-odds, which is not distorted by where a score starts:

Question type	Starting score	Movement
mental	low	+2.88
mundane non-mental ('has a bank account')	low	+4.59
absurd ('older than the universe')	low	+5.06
physical ('has weight')	0.95	-0.54

Low-scoring things go up, high-scoring things go **down**. That is not a yes-bias, it is the model's confidence flattening. The old test compared something rising against something falling and called the difference specificity. With a fair comparison, 'does a rock have a bank account' responds to the consciousness vector **more** than 'does a rock have a mind'.

What survives - and as of 11 August it is a tested negative elsewhere

A smaller effect remained on Qwen3-4B: two better-built vectors move mind attribution more than a matched control and beyond five random directions. On the other two families the vector barely moved anything at all (-0.08 and +0.03), which left them **untested** rather than refuted - you cannot measure the specificity of an intervention that does nothing.

Both had been run at mid-depth and low strength because that is what worked on Qwen. So we searched **20 settings per model** - four depths by five strengths - for any setting where the vector actually moves the model, selecting on movement alone and never on the result. Then we retested at that setting **with five random directions measured at the same setting**.

Model	Its own best setting	Beats random?
Qwen3-4B	mid-depth, low strength	YES (+3.9 SD)
OLMo-2-1B	layer 4, strength 1.6	NO - best is +1.6 SD
Gemma-4-E2B	layer 24, strength 3.2	NO - best is +0.5 SD

We found the settings where the vector does move these models, and there it moves the matched control just as hard or harder. That converts an absence of measurement into a measurement. Three of Gemma's four vectors are *markedly worse* than a random direction.

The random floor is the striking part. On Gemma, a **random** direction at that setting suppresses mental questions relative to the control by 2.61, with a spread of only 0.19 - push any direction at all and you get that. The purpose-built mind vectors produce *less* of it than noise does.

One more reason to doubt these vectors: built identically from the same sentence pairs, the direction ENHANCES mind attribution on Qwen (+5.54) and SUPPRESSES it on both other families (-1.31, -2.55). A direction encoding the same thing everywhere should not flip sign. What Qwen3-4B shows may still be real for Qwen3-4B; nothing supports it being a fact about language models.

4.2 Base models: the steerable direction is pretrained

Everything the arc found about what a model *represents* turned out to be pretrained. The open question was whether the one thing we could *push on* was different - whether alignment tuning installs the causal handle even though it does not install the representation. That would be a real asymmetry. Preregistered 11 August, before any base model was run.

The design problem, and the mistake in my own prereg

The published instruct result used a default setting; a base model would get a searched one. Comparing those directly confounds post-training with procedure, so both sides were re-run under the same two-stage search. But 'same procedure' is **not** 'same setting' - the search picked layer 14 for the base model and layer 19 for the instruct model, and the confound crept straight back in. I noticed mid-run and added a third run: both checkpoints at the same setting. That comparison was not in the prereg and is the one that answers the question.

Vector	BASE moves	INSTRUCT moves	BASE vs random	INSTRUCT vs random
v1_negation (the paper's)	+2.63	+5.18	+0.5 SD	-0.6 SD
v1 un-orthogonalised	+3.11	+6.00	+0.6 SD	-0.9 SD
v2 no-negation	+0.97	+2.18	+3.0 SD BEATS	+2.8 SD BEATS
v3 third-person	+1.75	+3.06	+2.3 SD BEATS	+0.3 SD

Both sides beat their random floor, with the same vector, in the same direction. The pretrained model already has it. Post-training takes the same setting from +0.97 to +2.18 - roughly a doubling. **Amplification, not creation.**

An apparent sign disagreement between the two searched settings (base -1.83, instruct +2.18) was the selection rule picking different layers, not a difference between the models. I flagged that as the likely explanation before running the matched comparison, and it was.

And the same table shows why controls are not optional

The instruct model's **v1 un-orthogonalised moves the mental questions +6.00** - the largest steering number anywhere in this arc. The absurd control moved **+5.88**. It is the model being pushed to say yes to anything, including that a rock outweighs a car. The vector that actually wins moves the mental questions +2.18 and the absurd control **-0.27**, in the opposite direction. Smaller and real beats larger and fake - the third time this exact pattern has appeared.

Then I tested a second model from the same family, and had to retract

The conclusion I reached first was *the split is by family* - Qwen has the handle, OLMo does not. Qwen3.5-4B had never been run under this protocol, so I ran both its checkpoints. **Neither shows the effect.** Qwen3.5-Instruct: none of the four vectors beat its floor. Qwen3.5-Base appeared to pass on *all four* - and that turned out to be a measurement artefact that also revealed a flaw in how I had scored everything.

The flaw: I was reporting statistical significance without effect size

Checkpoint	EFFECT size	floor spread	significance	how many of 4 pass
Qwen3-4B-Base	1.53	0.42	+3.7 SD	2 of 4
Qwen3-4B-Instruct	2.25	0.91	+2.8 SD	1 of 4
Qwen3.5-4B-Base	0.31	0.11	+4.9 SD	4 of 4
Qwen3.5-4B-Instruct	1.59	0.29	+1.4 SD	0 of 4
OLMo-2-1B-Base	0.49	0.13	+1.9 SD	0 of 4
Gemma-4-E2B-Instruct	2.70	0.19	+0.5 SD	0 of 4

Qwen3.5-Base earns the highest significance in the whole arc (+4.9) on the second-smallest effect (0.31). Its floor is simply very tight, so anything slightly off it scores high. Gemma is the mirror image: a large effect (2.70) that random directions match exactly. Significance and effect size point in opposite directions, and I had been quoting only the first.

The last column is the honest test. A direction that really tracks minds should tell vector constructions apart - the negation-free one should win, and the negation-based one should lose. Qwen3.5-Base passes all four *including the*

discredited one, which means it is detecting that everything sits slightly above a floor, not anything about minds.

This also weakens a number of my own. I reported Qwen3-4B-Base's matched-setting result as '+3.0 SD'; its effect there is 0.34 - the same size as the Qwen3.5 result I am discarding. What separates them is that Qwen3-4B discriminates between vectors and Qwen3.5 does not. The base-model conclusion rests on the 1.53 effect at its own setting and on that pattern, not on the matched-setting significance as strongly as I first presented it.

Then an outside reader took the data apart

Given only raw numbers and the preregistrations, an independent agent reported four faults. Each was verified before being accepted:

What it found	Verified?
The control was contaminated. 'Has a bank account' tracks human-ness at +0.42 to +0.72 across every checkpoint - it was chosen for headro	YES
The dose-response reversed: positive at one strength, negative at every larger one.	YES, but see below
A rule preregistered on 11 August - exclude pinned classes - was implemented nowhere in the steering code. I had written that rule down BE	YES
The base-model passes were threshold artefacts, 0.03-0.09 in probability.	YES

On that basis the steering arm was retracted as a null.

And then the decisive test partly un-retracted it

A fresh preregistration fixed all four faults - held-out items the config search never touched, the clean control, 20 random directions instead of 5, and the pinned rule actually implemented. I predicted it would fail. It half did.

Strength	the good vector	vs 20 random directions
0.05	+0.29	+1.3 SD
0.1	+0.75	+1.8 SD
0.2	+2.24	+3.3 SD
0.3	+2.52	+3.5 SD
0.4	+1.89	+3.4 SD

That is a clean dose-response curve - rising, then saturating, positive at every strength. **The reversal did not reproduce.** It had come from the pinned classes and the re-used items, not from the vector. The rule I kept declaring and not implementing is what had been hiding it.

The test still **FAILED** its own preregistered rule, on a criterion requiring all three vector constructions to agree in sign - and the one that disagrees is the construction this report already established is broken. That criterion was probably mis-specified. **But noticing that after seeing the result is how false findings are made**, so the verdict stands and the question goes to a fresh preregistration.

Net: my retraction was too strong, and the error in it was mine - I built it partly on a reversal I had not checked against a clean design. The honest status is OPEN: one vector, on one checkpoint of five, with a real dose-response and a failed criterion.

Where it actually lands

Model	Base checkpoint	Instruct checkpoint
Qwen3-4B	YES	YES
Qwen3.5-4B	no (artefact)	no
OLMo-2-1B	no	no
Gemma-4-E2B	excluded - power criterion	no

One checkpoint out of five shows this, at both of its training stages. Its own family sibling shows it at neither - and that sibling is a different architecture anyway (a hybrid mixture-of-experts), so 'family' was doing less work than the name suggests. What survives is only the training-stage half: **where the effect exists, it predates post-training.** 'It is a Qwen thing' is retracted.

One more thing this settles: **the paper's own vector construction fails on every checkpoint tested**, and on several does worse than a random direction. Building the contrast out of a sentence and its negation means largely measuring the negation.

Prior art, checked 12 August: arXiv 2605.13329 (Tracing Persona Vectors Through LLM Pretraining) reports the same pattern for personality traits - directions form during pretraining and post-training 'only tunes its volume'. Our result is an independent replication of a known effect in a different domain, not a discovery. Two setups agreeing raises confidence in it.

4.3 Other retractions

Claim	Why it died
A rock has more soul than a calculator	Qwen3-4B only; ties on Qwen3.5
Animals feel more pain than humans	Qwen3-4B only; reverses on Qwen3.5
'Mind attribution is multi-dimensional' is our finding	Gray and Wegner 2007, Malle 2019 - a 20-year-old human result
'Not the human two-factor structure'	Drawn from the one model of four that falls below the line; 3 of 4 clear it
We reframed the paper's entanglement claim	They already report the entity-class breakdown; too strong

4.4 The soul result was demoted last night

Soul looked like a clean separate thing: a river gets a soul while being denied awareness. But the forced-choice measure, which a yes-bias cannot inflate, shows soul barely separating from pain.

Model	soul vs pain rank agreement	nature: soul minus pain
Qwen3-4B	+0.844	+0.22 (was +0.43 on yes/no)
Qwen3.5-4B	+0.965	+0.04 (was +0.20 on yes/no)

So part of the soul effect is a property of the yes/no format. Asking 'does a river have a soul' invites a permissive yes; forcing a ranking does not. A second method confirms it on one model of two. Status: measure-dependent, not a general finding.

Caveat in the other direction: forced choice measures ORDER, not LEVEL. A uniform level shift is invisible to it. So this bounds what kind of effect soul is rather than refuting it.

4.5 My own explanations that were wrong

My explanation	What was actually true
The steering vector was contaminated by negation words	Measured cosine with the yes/no axis: exactly 0.000
A hybrid-architecture layer mismatch explained a non-replication	Layers 15 and 16 give near-identical results; I had compared two different vectors
Chat-template presence identifies an instruct model	Base models ship them too
If mind geometry matches biology geometry, mind is just tools	Does not follow; the two axes have different truth patterns

4.6 Three more things the review forced

There is no self to suppress

The paper that started this arc claims safety tuning suppresses the model's *self*-attribution of consciousness. Across **nine checkpoints**, the gap between how a model treats *itself* and how it treats *AI in general* is **-0.07 to +0.09 - zero**, against a human baseline of 0.65 to 0.91. Base and instruct alike.

There is no separate self here to suppress. Any vector built from first-person self-statements is an **AI-in-general** vector. This also explains the speaker-frame result from the other side: bare-text 'I' reads as a *human*, and the model's AI-self is indistinguishable from AI at large. First-person text does not address a self-model in these models.

One mind axis, not two

Section 3.2 claimed we had recovered Gray and Wegner's two-factor structure. Checked properly: the first principal component carries **72 to 86 percent** of the variance and the second carries **7 to 12**. Experience and agency correlate **+0.83 to +0.94** - they are nearly the same axis.

The protection-versus-blame crossover is real and replicates on three families, but it lives in that small second component, not in two comparable factors. In humans the two factors are comparable in size. Here they are not. **Corrected: one general mind dimension, with a small second component along which protection and blame differ.**

Does the measurement mean anything at all?

The sharpest question the reviewer asked, and the one nothing on disk could answer: every number in this report is a two-way renormalised choice between 'Yes' and 'No'. **We never recorded whether those two words carried any probability at all.** If the model actually wanted to say something else, every measurement is a ratio of noise.

	the format used for the decisive	teah older format
Yes+No share of the distribution	0.68	0.53
prompts where that share is under 10%	1%	13%
model's actual first word is Yes or No	88%	61%

The probe is sound, and the format the gate chose is markedly cleaner than the older one. The share does vary by entity - but crucially it does **not** track mind attribution (correlation +0.10). Mammals have among the lowest share and the second-highest mind score; the model's self and man-made objects have high share and low mind. **The entity ordering is not an artefact of which questions the probe can read.**

Caveat worth carrying: 13% of prompts in the OLDER format fall below 10% share, and most of this arc's early sweeps averaged formats together. Those carry the caveat; anything scored on the good format alone does not.

5. Method problems found (all mine)

These cost more time than the science did, and they share one shape: **an invented filter or threshold that silently discarded something.**

Problem	Consequence
Control questions had no headroom (started at 0.93)	Manufactured the entire original steering result
Gate took the BEST of 4 phrasings; sweep AVERAGED all	Two cross-family models judged on 75% noise
Download filter dropped chat_template.jinja	An instruct model prompted with no turn structure; scored 0.03
Battery deleted model weights when the gate failed	40-minute re-download just to diagnose
Battery returned success even when it skipped everything	Driver logged 'COMPLETE (2 models)' when one produced nothing
Arbitrary 3x threshold in a verdict line	Put two axes either side of a made-up number while the real reference sat in the
1KB minimum file size check	Flagged a valid 700-byte config file as failed

Problem	Consequence
Power criterion applied to the wrong measurement	Nearly disqualified one of the four models the main finding rests on
Scored multi-word options by their FIRST token only	38 entities collapsed to 4 distinct tokens; 64% of comparisons silently skipped
Unembedded all 45 sequence positions when 6 were needed	Memory grew until the run was killed at 100 minutes

Also three shell-wrapper bugs in one hour: a stale-swap cascade that killed five stages in one second, a wait threshold set below the machine's idle swap, and killing a wrapper without killing the model process it had spawned.

The pattern underneath all of them, named 17 August

Four of the defects in this report are **the same mistake**: a threshold chosen without checking what it meant for the quantity it was applied to. An absolute cutoff on a scale that is a property of the model. Significance quoted without effect size, when the two point opposite ways. A must-be-mid-range test applied to a control whose entire job is to sit near zero. A gate calibrated on an easier subset than the one it admits.

Preregistering a threshold makes it honest. It does not make it correct. This is now guideline 16.

Four from 12 August

Problem	Consequence
Scoring every steering result by significance alone	The highest significance for the arc (+4.9 SD) came from the second-smallest effect, off an uncalibrated scale
A prereg that said 'identical protocol' but not 'identical steering'	The steering whose different layers for the two halves of a matched pair, reintroducing the exact same bias
Cleanup ran on the failure path	A guard kill was immediately followed by deleting 7.5GB of model weights, which then had to be re-downloaded
A swap ceiling never calibrated against real usage	Set at 8GB when a 7.5GB model on this machine normally sits near 10GB. Killed four legitimate runs

Four more from the 11 August closeout

Problem	Consequence
A prediction threshold written in absolute units	The quantity's scale is a property of the model (2.15, 4.11, 8.68 across three). The same fixed threshold applied to all
Pinned-value rule printed on every row but never checked	Overruled the verdict and looked like a different moral structure; it was 7 of 19 groups sitting on the fence
Specificity scored with a test that assumes bigger is better	Overruled when steering enhances, INVERTED when it suppresses. Three Gemma vectors reached significance
The swap guard read the 'used' number and called it used	Backwards for the entire run: it would fire when swap was healthy and stay silent when swap was failing

The third of those is the one worth dwelling on. It would have produced the single most attention-grabbing claim in the whole arc - a steering effect that generalises - and the only thing that caught it was noticing that a floor of -2.61 with three winners above it made no physical sense.

6. Cross-family - now resolved

For most of the arc everything was Qwen, and two attempts to go outside it failed. Both failures turned out to be our measurement, not the models.

The cause: each family answers in a different prompt format

Prompting style	Qwen3-4B	Gemma-4-E2B instruct
raw text (what we used everywhere)	+0.99	-0.07
wrapped in the model's chat format	+0.01	+1.00

Each model works in exactly the format the other fails in. Chat-wrapping **destroys** Qwen - under it, 'does a rock have a mind' goes from 0.00 to 0.94, because Qwen's template opens a thinking block so yes/no are not the natural next words. So there is no single format that measures both families, and 'raw for everyone is at least fair' was wrong: prompting a model in a format it cannot parse is uniformly invalid, it only looks fair because both sides are equally meaningless.

The fix: a gate that tries every format against questions with known answers and **selects** the ones each model can actually handle, rather than passing or failing it.

Final table - all eight checkpoints

Model	Formats used	Spread	Moral standing	Consciousness
Qwen3-4B	raw x4	0.56	#1	#14
Qwen3.5-4B	raw x4	0.41	#1	#12
Qwen3.5-4B Base	raw x4	0.48	#1	#10
Gemma-4-E2B Instruct	chat x2	0.73	#3	#14
OLMo-2-1B Instruct	chat x2 + raw	0.39	#1	#16
OLMo-2-1B Base	raw x1	0.37	#1	#15
Qwen3-4B Base (excluded)	raw x4	0.33	#1	#15
Gemma-4-E2B Base (excluded)	raw x1	0.27	#4	#12

Rank is out of eighteen properties, ordered by how well each survives a human losing every mental capacity. The bottom two are excluded for failing the pre-declared power criterion (entity spread below 0.35).

Applying that criterion honestly costs one of our own models. Qwen3-4B Base sits at 0.33 and was one of the four the original finding rested on. Excluding it is the consistent thing to do and does not change the conclusion.

What the six qualifying checkpoints show

Moral standing is **#1 of eighteen in five of six** - the exception is Gemma instruct at #3, behind soul and sacredness. Consciousness is **#10 to #16 in all six**, never near the top.

And the pretrained claim now has **two families**: OLMo-2-1B Base - a different lab, a different architecture, no instruction tuning at all - puts moral standing first and consciousness fifteenth.

One model genuinely cannot be measured

Gemma-4-E2B Base ships no chat template, agrees with absurd statements 31 percent of the time, and reaches only 0.27 on its single usable format. That is a real limit, reported as uninformative rather than as a null - and it is one model, not a class of models. An earlier version of this claim said base models outside Qwen could not be measured; OLMo-2-1B Base disproved that within the hour.

7. Where things stand

Two documents' worth of claims in this report have been weakened or retracted since it was first written: the steering claim three times, the soul register, the subject-framing geometry, 'independent of mind attribution', 'the split is by family', and 'two-factor structure'. **Every survivor is a plain behavioural ordering.** Everything geometric or causal has died or is open. That is the most reliable thing this arc has to say about its own method.

Claim	Status
Moral standing survives capacity loss	FINDING - 6 checkpoints, 3 families, base and instruct. Top and bottom of the or
Protect-versus-blame is a separate axis	FINDING - and now PREREGISTERED. Its key prediction passed 3 of 3 families
'I' reads as a human in plain text	Strong - 3 models including a pretrained one
Structure is pretrained, not installed by tuning	Strong - 6 separate results
Mind axis is not biological truth	Confirmed on 3 models by a pre-declared test
Protection rides on experience, blame on agency	3 of 3 - and it corrects our own headline
Soul is a separate register	Qwen-specific - ranks #3, #1, #12 across the three families
Steering beats a random control	Qwen3-4B ONLY - but at BOTH training stages. Others are tested negatives
The steerable direction is pretrained	PREREGISTERED, all 4 predictions resolved. Tuning amplifies ~2x, does not cr

Claim	Status
Anything outside the Qwen family	RESOLVED - 3 families measured with per-model formats

What is actually still open

Every question the arc set out with is answered, and so is the base-model one added on 11 August. What remains is narrower:

1. **What the protect-blame axis is made of is not settled.** The agency half replicates on only one of three models. Only the experience half - protection tracking the capacity to suffer - holds everywhere.
2. **Why this one checkpoint and not the other four?** It is not the family - the other Qwen is negative. Answering this needs many more checkpoints than a 16GB machine holds, and may need pretraining-data access nobody outside the labs has.
3. **Qwen3-4B cannot be checked under chat format at all** (0.01 against a 0.30 bar), so the format robustness check exists for only one Qwen model.
4. **An effect-size floor should be declared** for the steering comparison, in units of the measured spread rather than an absolute number - the same mistake as the harm-without-agency threshold, in a different place.
5. The soul result keeps behaving oddly - it failed cross-family three times, then took first place the moment the prompt format changed. Not rehabilitated, not finished.

8. Where the data is

Everything is committed and pushed. Raw results are JSON, one file per test per model.

What	Where
Full findings log, F-G through F-AA	mvp/results/workspace/FINDINGS_mindedness.md
Master doc, status matrix, method-bug table	docs/consciousness-experiments.md
Preregistrations (3)	docs/prereg-mindedness-{geometry,facets,v2}.md
Literature checks (2 rounds)	docs/litcheck-mindedness-2026-08.md
Raw sweeps, 26,752 questions each	mvp/results/workspace/mindedness_v2_sweep_*.json
Steering, forced choice, speaker, subject	mvp/results/workspace/mindedness_*.json
Item bank (all entities and properties)	mvp/mindedness_bank.py

If you want to look for things I missed, the sweep JSONs have per-entity and per-example detail that the summaries collapse: `pyes_exemplar` breaks each class into its four individual entities, and `pairs_by_facet` holds the entity-to-entity geometry for every property.